

ChE 596 Polymer Rheology - Fall 2024

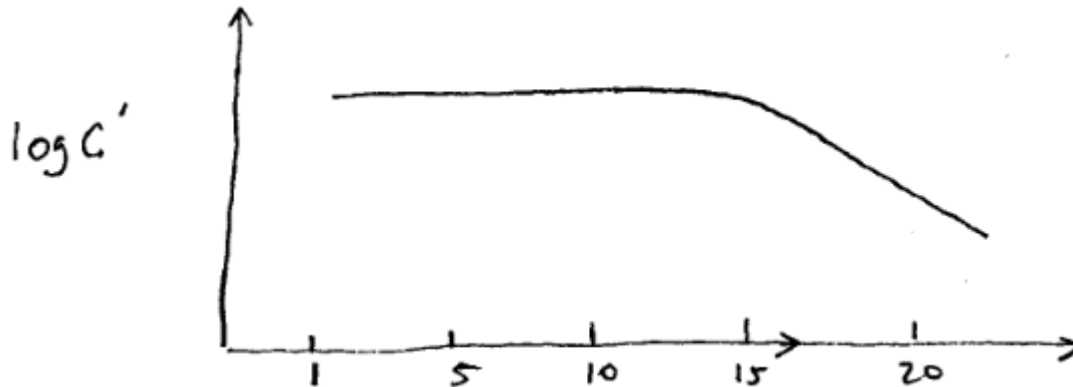
Homework 3

Due date: Tuesday, October 8th at 1:30 PM EDT

Show all your work for full credit.

Problem 1

A typical plot of the elastic modulus of a polymer as a function of strain is shown below. For what strain limits will the linear viscoelastic model be valid for this material?



Problem 2

Your friend has carefully calculated the stress tensor for a complex flow and come up with the following total stress tensor.

$$\underline{\underline{T}} = \begin{bmatrix} 20 & 8 & 0 \\ 8 & 15 & 3 \\ 0 & 3 & 15 \end{bmatrix}$$

- (a) Assuming that the fluid is Newtonian, comment on the nature of the flow
- (b) Assuming that the fluid is viscoelastic, comment on the nature of the flow.

Problem 3

You are trying to check the reproducibility of your lab experiments by repeating them. You are working with a polymer which has a longest relaxation time constant of 1000 seconds. To check for reproducibility, you are measuring the viscosity-shear rate behavior by doing experiments back to back. Unfortunately, your results are not overlapping. What could be wrong and what would you do to rectify it, if possible?

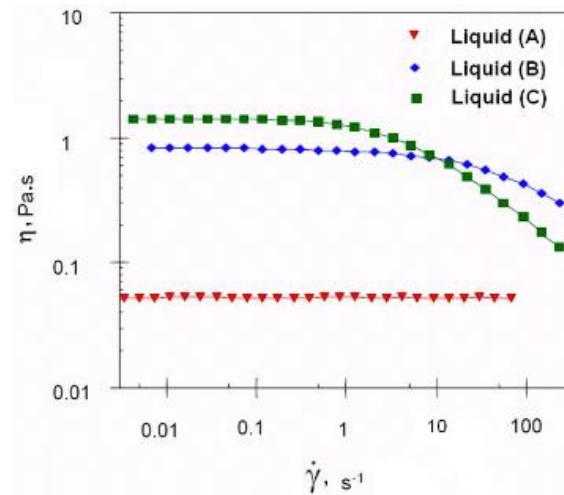
Problem 4

Consider a small amplitude oscillatory shear flow being modeled using the linear viscoelastic equation with one exponential element, i.e., $G(t-t') = \eta/\lambda \exp[-(t-t')/\lambda]$. What would η' and η'' be or how would they each scale with frequency ω as: (a) $\lambda \rightarrow 0$; (b) $\omega \rightarrow 0$.

Problem 5

Steady shear viscosity versus shear rate data is provided below for materials on a log-log scale

- (a) Estimate the zero shear viscosity for each liquid
- (b) Estimate the relaxation time for each liquid. Explain how you got them, if needed
- (c) Assuming that all of them are the same material, comment on the molecular weight and its distribution for each liquid/melt (e.g., A has the medium mol weight and most polydispersity, etc)



Problem 6

- (a) Your supervisor at No-Rheology-Background Technologies has given you a material and asked you to measure its viscosity at a specified temperature. You tried the cone and plate geometry, but the transducer did not pick up any signal. Then you used the typical couette geometry but still the noise was too high as the torque signal was weak, especially at low shear rates.
- What other geometry (or geometries) may enable you to measure the viscosity of such a liquid?
 - Are there other strategies you could take to get adequate torque signal?
- (b) Your supervisor (again!) comes back and asks you to measure the viscosity of another material for a range of temperatures from 25 °C to 150 °C. What geometry would you use in this case and why? Also describe how you would go about doing the experiments.